

MDI3003
ADVANCED PREDICTIVE ANALYTICS
LAB-05
PRODUCT AND BRAND SENTIMENT PREDICTION
FROM TWEET DATA

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COURSE TITLE : Advanced Predictive Analytics

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CONTENTS

1. Executive Summary

2. Intended Use and Scope

3. Dataset and Provenance

4. Data Quality

5. Methods

6. Model Development

7. Final Evaluation

8. Interpretation

9. Subgroup and Robustness

10. Limitations and Risk

11. Deployment Considerations

12. Conclusion

References

1. Executive Summary

This laboratory study evaluates supervised sentiment classification for short, noisy social-media text using two three-class tweet datasets: the TweetEval sentiment benchmark and the Twitter US Airline Sentiment dataset. The experimental objective was to determine how far transparent sparse-text methods can go on tweet sentiment, and then measure whether sequence modeling and a Twitter-domain pretrained Transformer provide a meaningful improvement when evaluated on the same frozen test partitions. The analysis therefore progresses from Dummy and VADER baselines through TF-IDF with Multinomial Naive Bayes, Logistic Regression and LinearSVC, followed by character TF-IDF, a trainable embedding-based BiLSTM and Twitter-RoBERTa.

The two datasets expose an important domain effect. TweetEval contains 59,899 tweets across its official train, validation and test partitions, with negative, neutral and positive labels. The training corpus is strongly skewed toward neutral tweets. The airline dataset contains 14,640 tweets and is substantially negative-heavy. The same negative/neutral/positive label semantics were used for both datasets, allowing a documented cross-dataset transfer experiment without inventing a neutral class for a binary corpus.

For the classical core, LinearSVC was selected as the learned sparse-text model using validation evidence. Its frozen test Macro F1 was 0.5684 on TweetEval and 0.7468 on the airline dataset. Character TF-IDF improved TweetEval's test Macro F1 to 0.5900 and produced a small test improvement over word TF-IDF on Dataset 1; on Dataset 2 it reached 0.7457, essentially matching the word representation while having a slightly higher cross-validation Macro F1.

The advanced results show a much larger benefit from Twitter-domain contextual pretraining than from the BiLSTM. Twitter-RoBERTa achieved Macro F1 0.7188 and accuracy 0.7174 on TweetEval, versus 0.5684 Macro F1 for LinearSVC. On the airline dataset it achieved Macro F1 0.8157 and accuracy 0.8593, versus 0.7468 and 0.8128 respectively for LinearSVC. The corresponding gains in Macro F1 are approximately 0.1504 and 0.0689. BiLSTM reached Macro F1 0.5850 on Dataset 1 and 0.7510 on Dataset 2, so it did not close the gap to the Twitter-specific Transformer.

The main practical conclusion is not simply that the largest model is best. Twitter-RoBERTa required about 1,079.5 s of training and 22.84 s of test inference on Dataset 1, and 355.2 s and 4.61 s respectively on Dataset 2, with roughly 124.6 million parameters and a checkpoint size near 2.86 GB. LinearSVC required only 7.3 s and 0.617 s on Dataset 1 and 0.7 s and 0.059 s on Dataset 2. Thus the Transformer is justified when the additional predictive quality matters enough to pay for substantially higher compute, memory and reproducibility cost.

The experiments also demonstrate that robustness is materially affected by domain and time. LinearSVC transfer from the airline domain to TweetEval fell to Macro F1 0.4246 in the Dataset 2 notebook's transfer run, while the reverse direction produced Macro F1 0.5338. A separate temporal holdout on the airline data reduced LinearSVC Macro F1 from 0.7468 under the random stratified split to 0.7138, a decrease of 0.0330. These results support treating sentiment models as domain- and time-dependent analytical instruments rather than universal measures of public opinion.

2. Intended Use and Scope

2.1 Analytical objective

The system addresses a supervised three-class classification problem: given a tweet, predict whether the expressed sentiment is negative, neutral or positive. Dataset 2 additionally supplies an airline/entity field, allowing the sentiment target to be interpreted explicitly in relation to a named service entity. Dataset 1 is a

general Twitter sentiment benchmark and therefore supports standardized sentiment modeling rather than a product-specific business claim.

2.2 Intended use

The intended use is aggregate analytical monitoring: summarizing sentiment distributions, identifying recurring areas of positive or negative discussion, comparing model behavior across classes, and supporting exploratory product or service feedback analysis. The output is appropriate for dashboards, research analysis, prioritization of topics for human investigation, and model-development experiments.

2.3 Non-intended use

The classifier is not intended to determine an individual's worth, eligibility, employment status, credit access, or other consequential outcome. A tweet-level prediction is not equivalent to a verified customer complaint, and aggregate tweet proportions are not estimates of the sentiment of an entire customer population. The data are historical and sampled from Twitter, so they can be affected by platform behavior, user composition, event timing and annotation conventions.

2.4 Scope of the completed work

The completed work covers data auditing, minimal tweet normalization, exploratory analysis, frozen train/validation/test evaluation, Dummy and VADER baselines, TF-IDF with MultinomialNB/Logistic Regression/LinearSVC, a controlled preprocessing comparison, character TF-IDF, BiLSTM training, Twitter-RoBERTa fine-tuning, multi-seed BiLSTM evaluation, confidence and abstention analysis, challenge-set heuristics, cross-dataset transfer, temporal holdout on Dataset 2, efficiency comparison, and save/reload checks for the classical and BiLSTM artifacts.

3. Dataset and Provenance

3.1 Dataset overview

Table 3.1 — Dataset provenance and experimental partitions

Dataset	Source	Total rows	Labels	Train / Val / Test	Domain
D1 – TweetEval Sentiment	cardiffnlp/tweet_eval, sentiment configuration	59,899	Negative / Neutral / Positive	45,615 / 2,000 / 12,284	General Twitter sentiment benchmark
D2 – Twitter US Airline Sentiment	CrowdFlower/Figure Eight dataset distributed through Kaggle; Tweets.csv	14,640	Negative / Neutral / Positive	10,248 / 2,196 / 2,196	Customer/service sentiment toward six airlines

The faculty manual identifies TweetEval as a standardized benchmark with fixed splits and the Twitter US Airline Sentiment dataset as the preferred product/entity-oriented core. The actual notebooks followed those two sources. TweetEval was loaded through its Hugging Face dataset configuration, preserving the official train/validation/test partitions. The airline notebook loaded Tweets.csv and created a reproducible 70/15/15 stratified split with random seed 42.

The airline dataset was collected in February 2015 and contains tweets concerning six U.S. airlines. The public dataset description states that contributors first assigned positive, negative or neutral sentiment and then

categorized negative reasons. This temporal and annotation context is important because the model is learning a historical service-discussion distribution rather than a current representation of airline customers.

3.2 Label harmonization

Table 3.2 — Cross-dataset label-harmonization record

Dataset	Original labels	Mapped labels	Compatible	Justification
TweetEval	negative / neutral / positive	negative / neutral / positive	Yes	Direct three-class correspondence
US Airline	negative / neutral / positive	negative / neutral / positive	Yes	Direct three-class business/service sentiment

The labels are semantically compatible at the polarity-class level, so cross-dataset transfer was performed only after recording the mapping. Compatibility does not imply identical domains: TweetEval represents heterogeneous Twitter sentiment, whereas the airline corpus is concentrated on service experiences. This distinction is central to interpreting transfer performance.

3.3 Class distributions

Table 3.3 — Dataset 1 class distribution

Class	Count	Share
neutral	20673	45.32%
positive	17849	39.13%
negative	7093	15.55%

Table 3.4 — Dataset 2 class distribution

Class	Count	Share
negative	9178	62.69%
neutral	3099	21.17%
positive	2363	16.14%

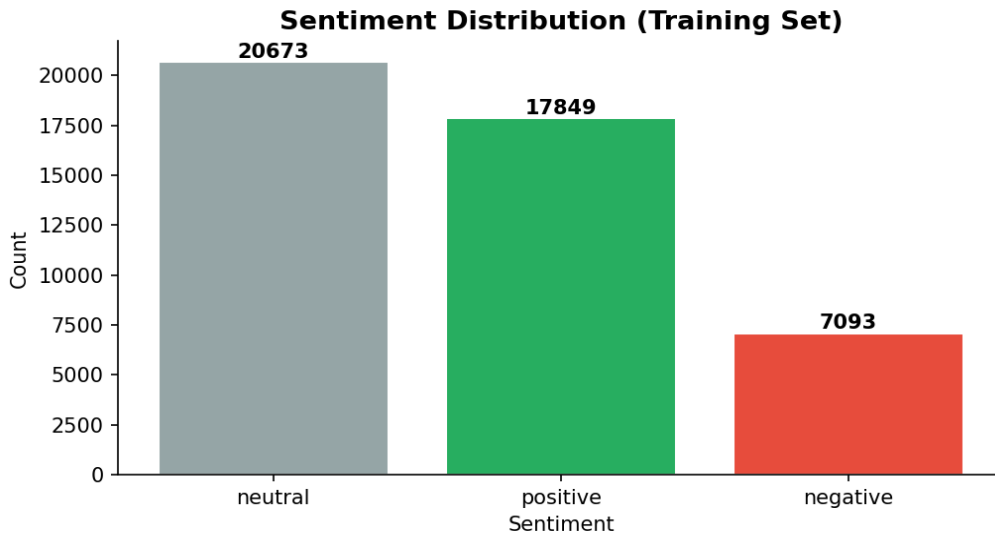


Figure 3.1 — TweetEval training-set sentiment distribution.

Dataset 1 contains 20,673 neutral tweets, 17,849 positive tweets and 7,093 negative tweets in the reported distribution. Neutral therefore represents about 45.3% of the corpus, while negative represents about 15.5%. This imbalance makes Macro F1 more informative than accuracy alone because a model can obtain a reasonable overall accuracy while under-serving the minority negative class.

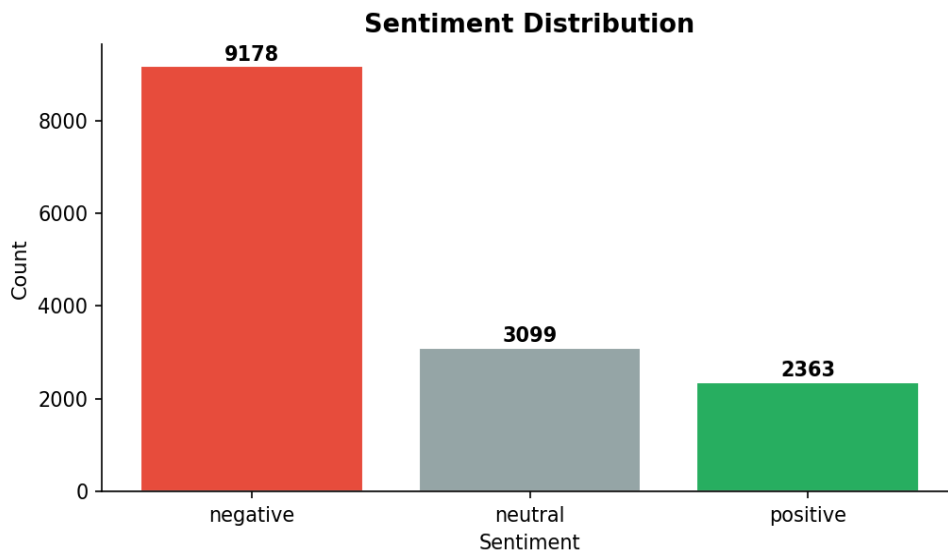


Figure 3.2 — Twitter US Airline Sentiment class distribution.

Dataset 2 is more strongly skewed: 9,178 negative tweets (62.7%), 3,099 neutral tweets (21.2%) and 2,363 positive tweets (16.1%). The negative majority makes weighted metrics more forgiving than Macro F1 and creates a strong reason to inspect per-class recall, particularly for neutral and positive sentiment.

4. Data Quality

4.1 Audit and governance

The notebooks explicitly audited dataset shape, schema, label values, missing values and duplicate text. Dataset 2 also checked duplicate rows and duplicate text and verified the presence of text, airline_sentiment and airline fields. Dataset 1 used the official benchmark splits and checked missing values and duplicate texts in the training partition. The persisted artifacts do not retain every printed audit count, so this report does not invent duplicate or missing-value totals that were not preserved in the output tables.

The modeling workflow excluded direct identifiers and did not use usernames, locations, coordinates or annotation-confidence fields as predictors. The manual specifically identifies sentiment-confidence and negative-reason metadata as leakage risks because they are tied to the target. The implemented classifiers used tweet text and, for the entity analysis, the airline field as contextual information rather than as a hidden target proxy.

4.2 Minimal preprocessing

The implemented cleaning rule is deliberately conservative. URLs are replaced with the token <URL>, mentions are replaced with <USER>, and repeated whitespace is normalized. The procedure preserves hashtags, emojis, negation and punctuation rather than stripping them by default. Empty cleaned records are removed. This design is consistent with the observation that tweet sentiment can be encoded in symbols, emphatic spelling, hashtags and short contextual expressions.

A controlled ablation compared the minimal representation with a more normalized representation using lowercasing and character stripping. On Dataset 1, the Logistic Regression Macro F1 changed from 0.5610 to 0.5638; on Dataset 2 it changed from 0.7122 to 0.7143. The small gains indicate that the additional normalization did not transform the task, supporting the use of the simpler preprocessing rule for the main experiments.

Table 4.1 — Dataset 1 preprocessing ablation

Preprocessing	Macro F1
Minimal (URL+mention replace)	0.5609931347716716
Normalized (lowercase+strip)	0.5638247063769491

Table 4.2 — Dataset 2 preprocessing ablation

Preprocessing	Macro F1
Minimal (URL+mention replace)	0.7121799927979793
Normalized (lowercase+strip)	0.714325818645845

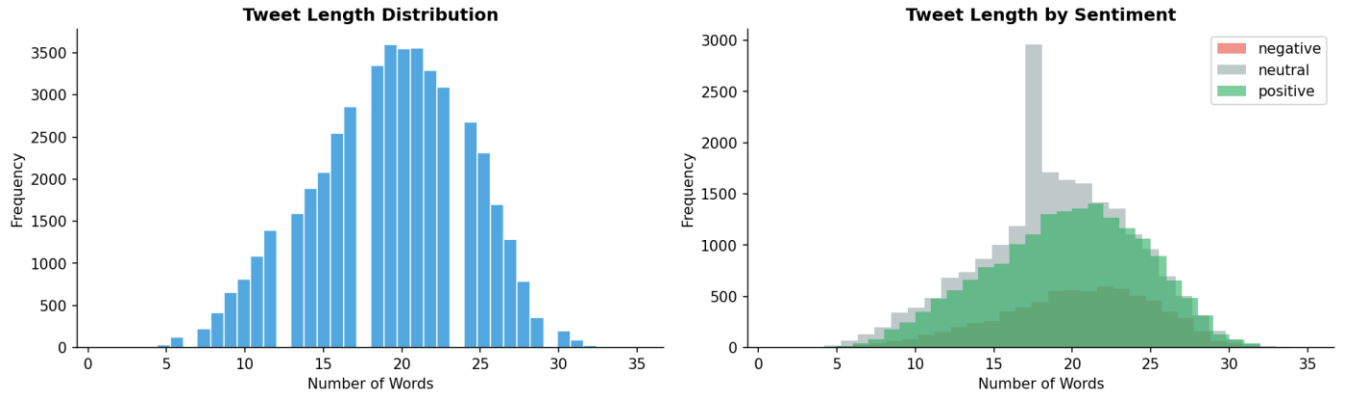


Figure 4.1 — Dataset 1 tweet-length distributions before modeling.

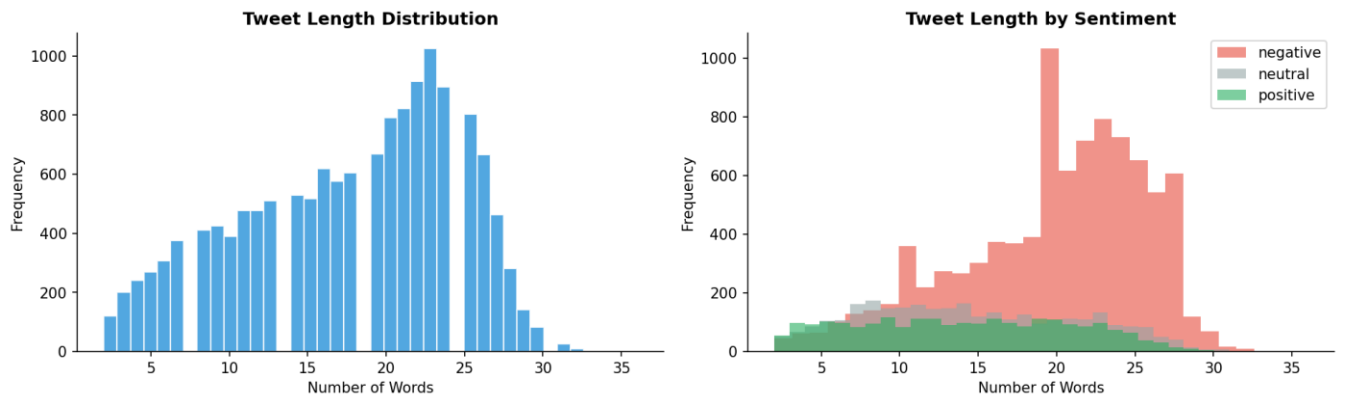


Figure 4.2 — Dataset 2 tweet-length distributions before modeling.

5. Methods

5.1 Problem formulation

Let x denote a cleaned tweet and $y \in \{\text{negative, neutral, positive}\}$ its dataset-defined sentiment label. The task is to learn a function $f(x)$ that minimizes classification error on unseen tweets while preserving performance across all sentiment classes. Because the class frequencies are unequal, Macro F1 is treated as the primary comparative metric, with accuracy, weighted F1 and per-class precision/recall used as complementary evidence.

5.2 Experimental workflow

The completed workflow is: dataset loading → schema and quality audit → minimal normalization → fixed split → exploratory analysis → Dummy and VADER baselines → word TF-IDF classical models → validation/CV model comparison → locked-test evaluation → error analysis → character TF-IDF → BiLSTM → Twitter-RoBERTa → uncertainty/abstention and robustness experiments → cross-dataset and temporal analysis → efficiency and reproducibility checks.

5.3 TF-IDF representation

For term t in document d , the implementation follows the standard TF-IDF idea: $\text{tfidf}(t,d) = \text{tf}(t,d) \times \log(N/\text{df}(t))$, where tf measures term occurrence and df measures the number of documents containing the term. Word unigrams and bigrams were used with sublinear term frequency, minimum document frequency of 2 and a maximum feature limit. Character n -grams of lengths 2–5 were evaluated as an additional representation. Character features can capture spelling variation, partial hashtags, morphology and noisy word forms that are common in social-media text.

5.4 Baselines

DummyClassifier provides a non-linguistic reference point by predicting the most frequent class. VADER provides a no-training lexicon/rule baseline designed for social-media language. VADER assigns positive, neutral or negative labels from its compound sentiment score using the implemented ± 0.05 thresholds.

5.5 Classical models

MultinomialNB estimates class-conditional token evidence under a conditional-independence assumption. Logistic Regression learns a linear decision function over sparse TF-IDF features and produces class probabilities. LinearSVC learns a large-margin linear separator and is well suited to high-dimensional sparse text. The notebooks use identical word TF-IDF pipelines for the three learned classical models so that the comparison isolates the classifier more cleanly.

5.6 BiLSTM

The BiLSTM converts training tweets into padded token sequences using a vocabulary fitted only on the training text. The model contains a trainable embedding layer, a Bidirectional LSTM with 64 units, dropout, a 64-unit ReLU dense layer and a three-class softmax output. Adam with learning rate 0.001 is used, with batch size 64 and early stopping on validation accuracy. The sequence length is the 95th percentile of training tweet lengths, which resulted in 27 tokens in both datasets.

5.7 Twitter-RoBERTa

The advanced Transformer is `cardiffnlp/twitter-roberta-base-sentiment-latest`. The fine-tuning setup uses maximum sequence length 128, batch size 16, learning rate 2×10^{-5} , weight decay 0.01 and three epochs, with validation-based checkpoint selection. The model contains 124,647,939 parameters and is therefore orders of magnitude larger than the BiLSTM and sparse linear models.

5.8 Evaluation methodology

Dataset 1 uses the official TweetEval train/validation/test partitions. Dataset 2 uses a fixed stratified 70/15/15 split with random seed 42. All comparative models use the same frozen test partition. Five-fold stratified cross-validation was performed on the pooled training and validation data for the classical models, leaving the final test set untouched. Model selection is therefore based on validation/CV evidence rather than final-test performance.

5.9 Metrics

Accuracy is the fraction of correct predictions. Precision measures how many predicted instances of a class are correct; recall measures how many true instances of a class are recovered. F1 is the harmonic mean of precision and recall. Macro F1 averages the three class F1 values equally, while weighted F1 weights them by support. Confusion matrices reveal the direction of errors, which is especially useful when neutral and positive tweets are confused with negative tweets.

6. Model Development

6.1 Classical development and validation

Table 6.1 — Dataset 1 five-fold cross-validation results

Model	CV Macro F1 Mean	CV Macro F1 SD	CV Weighted F1 Mean	CV Weighted F1 SD	CV Training Time (s)
MultinomialNB	0.5796	0.0043	0.6199	0.0035	13.695
Logistic Regression	0.6071	0.0025	0.6527	0.0038	28.793
LinearSVC	0.621	0.0037	0.6545	0.0033	14.937

Table 6.2 — Dataset 2 five-fold cross-validation results

Model	CV Macro F1 Mean	CV Macro F1 SD	CV Weighted F1 Mean	CV Weighted F1 SD	CV Training Time (s)
MultinomialNB	0.6943	0.007	0.7649	0.0061	4.205
Logistic Regression	0.7124	0.0117	0.779	0.0089	6.507
LinearSVC	0.7367	0.0086	0.796	0.0066	2.697

The cross-validation results show a consistent ordering among the three learned classical models. On Dataset 1, LinearSVC has the highest Macro F1 mean at 0.6210, followed by Logistic Regression at 0.6071 and MultinomialNB at 0.5796. On Dataset 2, LinearSVC again leads with 0.7367, ahead of Logistic Regression at 0.7124 and MultinomialNB at 0.6943. The relatively small standard deviations indicate that the ranking is not being driven by a single unusually favorable fold.

5-Fold CV Macro F1 (mean ± SD) — Dataset 1

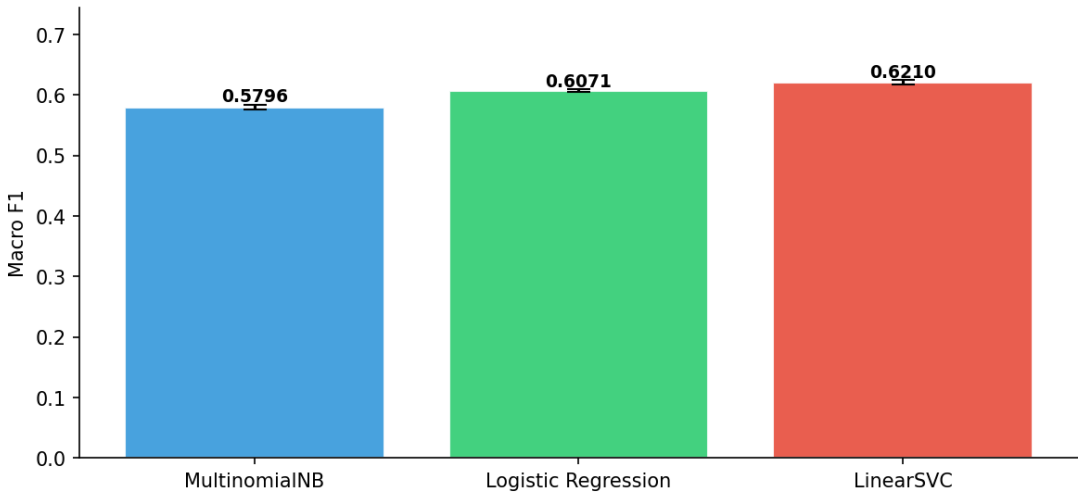


Figure 6.1 — Dataset 1 five-fold cross-validation Macro F1 comparison.

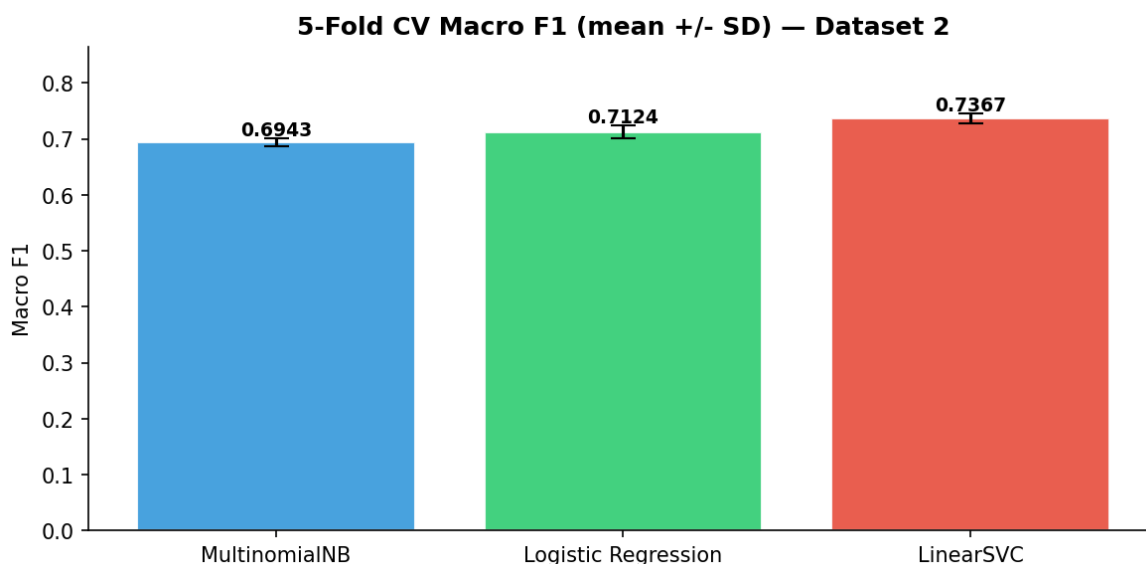


Figure 6.2 — Dataset 2 five-fold cross-validation Macro F1 comparison.

6.2 Character TF-IDF

Table 6.3 — Dataset 1 word versus character TF-IDF

Representation	CV Macro F1	CV SD	Test Macro F1	Test Weighted F1	Test Accuracy
Word TF-IDF (1-2gram)	0.621	0.0037	0.5684	0.5799	0.585
Char TF-IDF (2-5gram)	0.6386	0.0037	0.59	0.6025	0.6072

Table 6.4 — Dataset 2 word versus character TF-IDF

Representation	CV Macro F1	CV SD	Test Macro F1	Test Weighted F1	Test Accuracy
Word TF-IDF (1-2gram)	0.7367	0.0086	0.7468	0.8056	0.8128
Char TF-IDF (2-5gram)	0.7523	0.0061	0.7457	0.8062	0.8124

Character TF-IDF provided the clearest representation-level gain on Dataset 1: CV Macro F1 increased from 0.6210 to 0.6386 and test Macro F1 increased from 0.5684 to 0.5900. Dataset 2 showed a different pattern. Character TF-IDF had the stronger CV Macro F1 (0.7523 versus 0.7367) but a slightly lower test Macro F1 (0.7457 versus 0.7468). This is a useful example of why validation evidence and locked-test evidence should be kept separate.

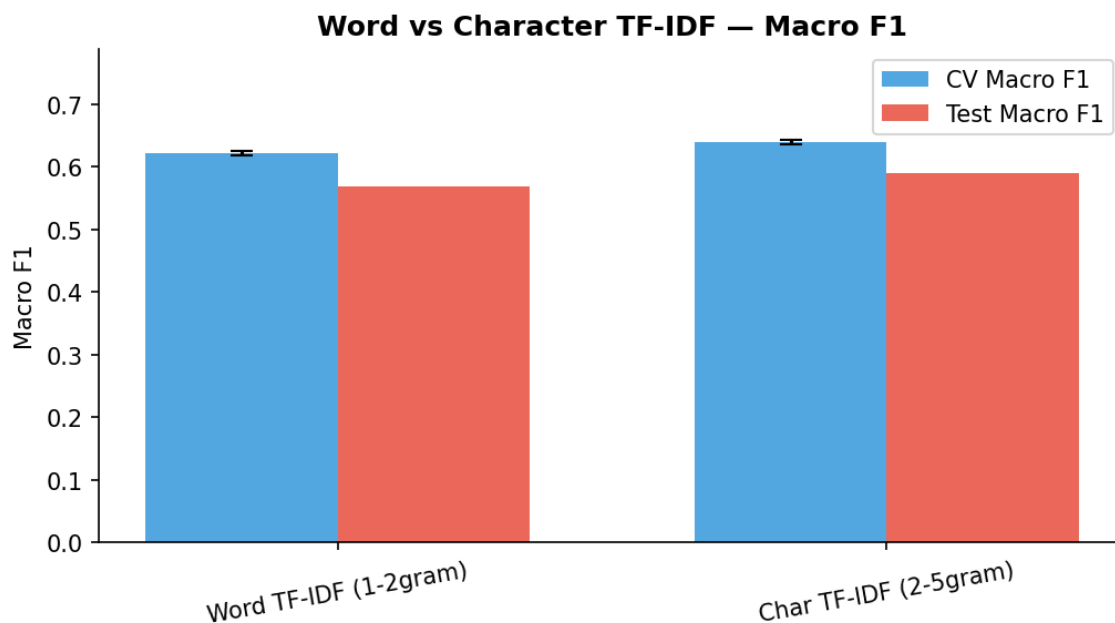


Figure 6.3 — Dataset 1 character TF-IDF comparison.

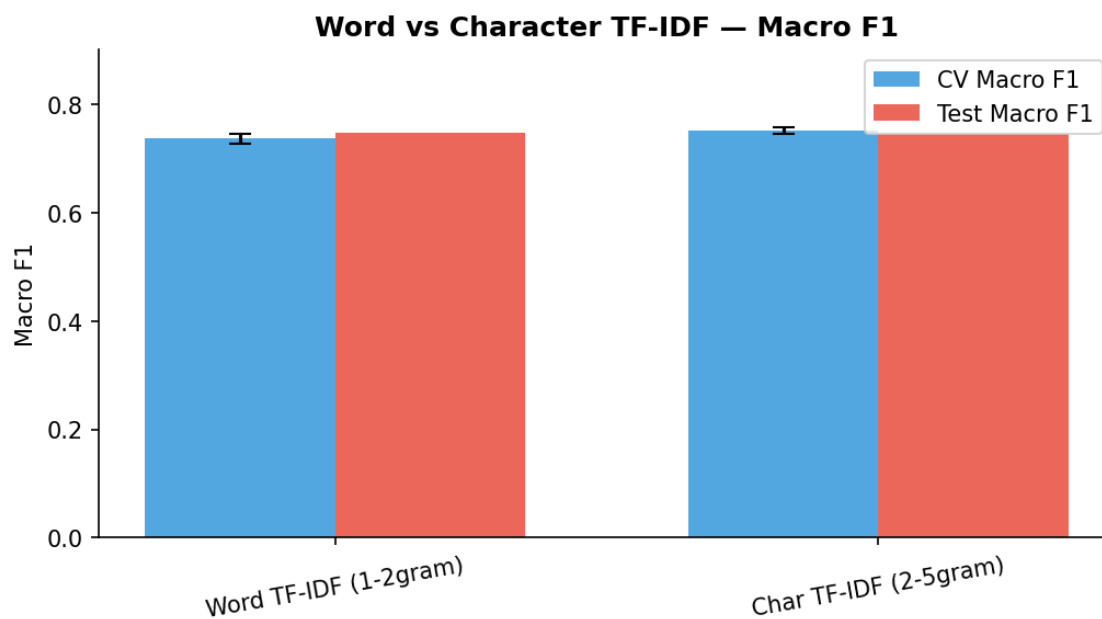


Figure 6.4 — Dataset 2 character TF-IDF comparison.

6.3 BiLSTM development

Table 6.5 — Implemented BiLSTM configurations

Dataset	Vocabulary	Embedding dim.	LSTM units	Dropout	Batch	Max sequence	Best epoch	Parameters	Model size (MB)
Dataset 1	20,000	128	64	0.3	64	27	2	2,667,267	30.57
Dataset 2	15,000	128	64	0.3	64	27	2	2,027,267	23.25

The BiLSTM reached its best validation point early, at epoch 2 in both datasets. This early stopping behavior is consistent with the short-text setting and the limited number of training epochs. The multi-seed results show that the model is not perfectly stable: Dataset 1 test Macro F1 values were 0.5850, 0.5896 and 0.5652 for seeds 42, 123 and 2026; Dataset 2 values were 0.7510, 0.7477 and 0.7316. Using the notebook's population standard deviation, the corresponding means are 0.5799 (SD 0.0106) for Dataset 1 and 0.7434 (SD 0.0085) for Dataset 2.

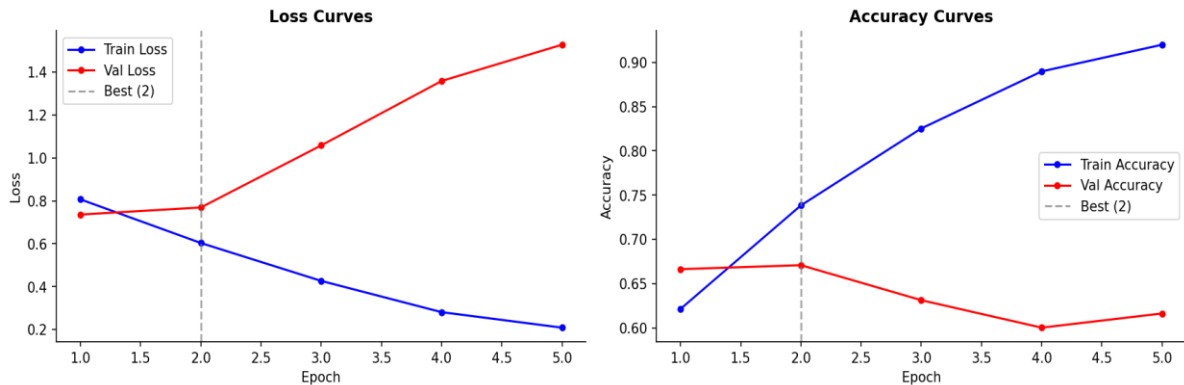


Figure 6.5 — Dataset 1 BiLSTM training and validation loss.

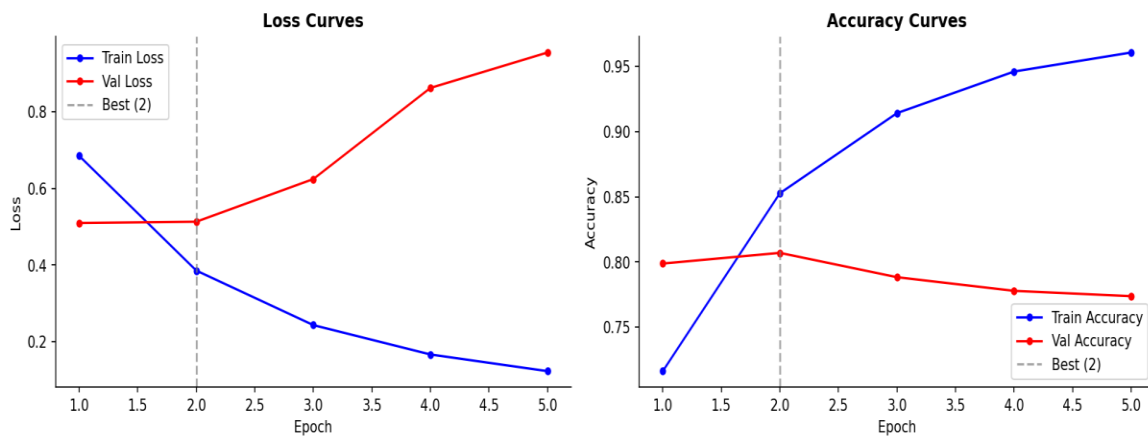


Figure 6.6 — Dataset 2 BiLSTM training and validation loss.

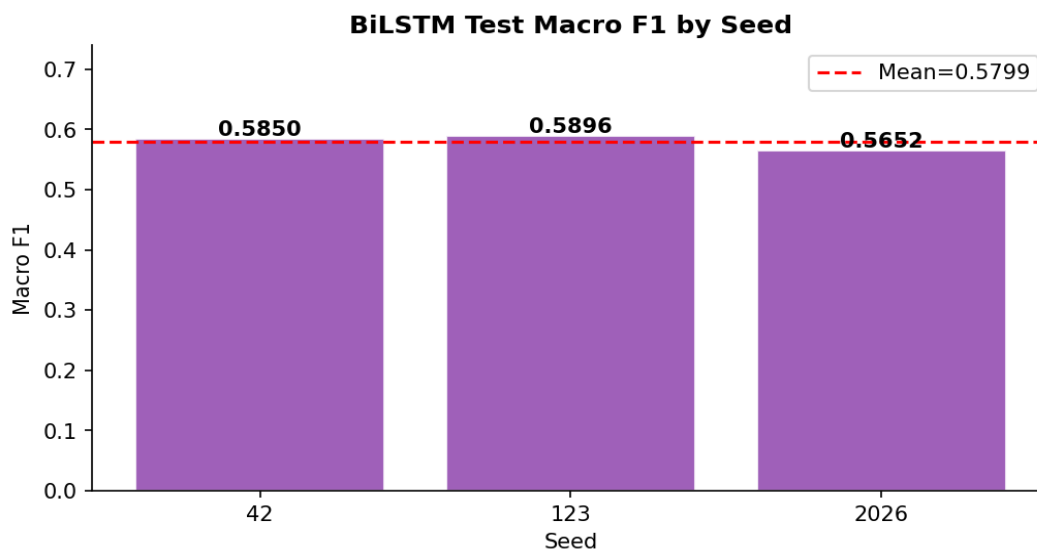


Figure 6.7 — Dataset 1 BiLSTM Macro F1 across completed seeds.

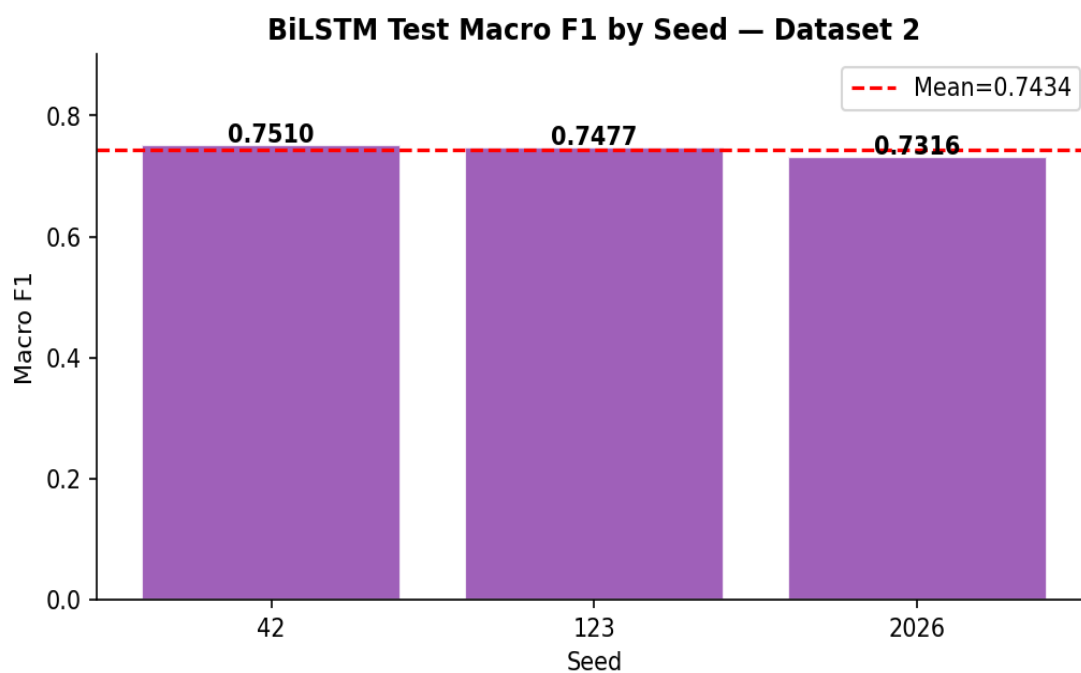


Figure 6.8 — Dataset 2 BiLSTM Macro F1 across completed seeds.

6.4 Twitter-RoBERTa development

Table 6.6 — Dataset 1 Twitter-RoBERTa fine-tuning result

Model	Model ID	Seed	Max Seq Len	Batch Size	Learning Rate	Weight Decay	Epochs	Parameters	Training Time (s)	Inference Time (s)	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Twitter-RoBERTa	cardiffnlp/twitter-roberta-base-sentiment-latest	42	128	16	2e-05	0.01	3	124647939	1079.5	22.84	0.7174	0.7092	0.7336	0.7188	0.7166

Table 6.7 — Dataset 2 Twitter-RoBERTa fine-tuning result

Model	Model ID	Seed	Max Seq Len	Batch Size	Learning Rate	Weight Decay	Epochs	Parameters	Model Size (MB)	Training Time (s)	Inference Time (s)	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Twitter-RoBERTa	cardiffnlp/twitter-roberta-base-sentiment-latest	42	128	16	2e-05	0.01	3	124647939	2860.19	355.2	4.61	0.8593	0.8125	0.819	0.8157	0.8602

The completed Transformer runs used seed 42. Dataset 1 required 1,079.5 seconds of training and 22.84 seconds for test inference. Dataset 2 required 355.2 seconds and 4.61 seconds. The discrepancy in runtime is consistent with the much larger Dataset 1 training partition. The model contains 124,647,939 parameters and the saved checkpoint is approximately 2.86 GB.

7. Final Evaluation

7.1 Overall test performance

Table 7.1 — Dataset 1 final advanced comparison

Model	Representation	Macro F1	Weighted F1	Accuracy	Training Time	Inference Latency	Parameters	Model Size (MB)
Logistic Regression	Word TF-IDF (1-2gram)	0.561	0.5759	0.593	29.0s	0.884s	nan	nan
LinearSVC	Word TF-IDF (1-2gram)	0.5684	0.5799	0.585	7.3s	0.617s	nan	nan
Char TF-IDF + LinearSVC	Char TF-IDF (2-5gram)	0.59	0.6025	0.6072	14.7s	2.070s	nan	nan
BiLSTM	Embedding + BiLSTM	0.585	0.5947	0.5948	104.0s	1.133s	2667267.0	30.57
Twitter-RoBERTa	Transformer (125M)	0.7188	0.7166	0.7174	1079.5s	22.844s	124647939.0	2860.23

Table 7.2 — Dataset 2 final advanced comparison

Model	Representation	Macro F1	Weighted F1	Accuracy	Training Time	Inference Latency	Parameters	Model Size (MB)
Logistic Regression	Word TF-IDF (1-2gram)	0.7122	0.7843	0.7983	3.7s	0.199s	nan	nan
LinearSVC	Word TF-IDF (1-2gram)	0.7468	0.8056	0.8128	0.7s	0.059s	nan	nan
Char TF-IDF + LinearSVC	Char TF-IDF (2-5gram)	0.7457	0.8062	0.8124	2.2s	0.378s	nan	nan
BiLSTM	Embedding + BiLSTM	0.751	0.8073	0.8097	20.4s	0.592s	2027267.0	23.25
Twitter-RoBERTa	Transformer (125M)	0.8157	0.8602	0.8593	355.2s	4.606s	124647939.0	2860.19

Twitter-RoBERTa is the strongest completed model on both datasets. On Dataset 1, its Macro F1 of 0.7188 exceeds LinearSVC's 0.5684 by 0.1504. On Dataset 2, its Macro F1 of 0.8157 exceeds LinearSVC's 0.7468 by 0.0689. The gain is larger on Dataset 1, where the general benchmark contains more heterogeneous Twitter language and the sparse models struggle more.

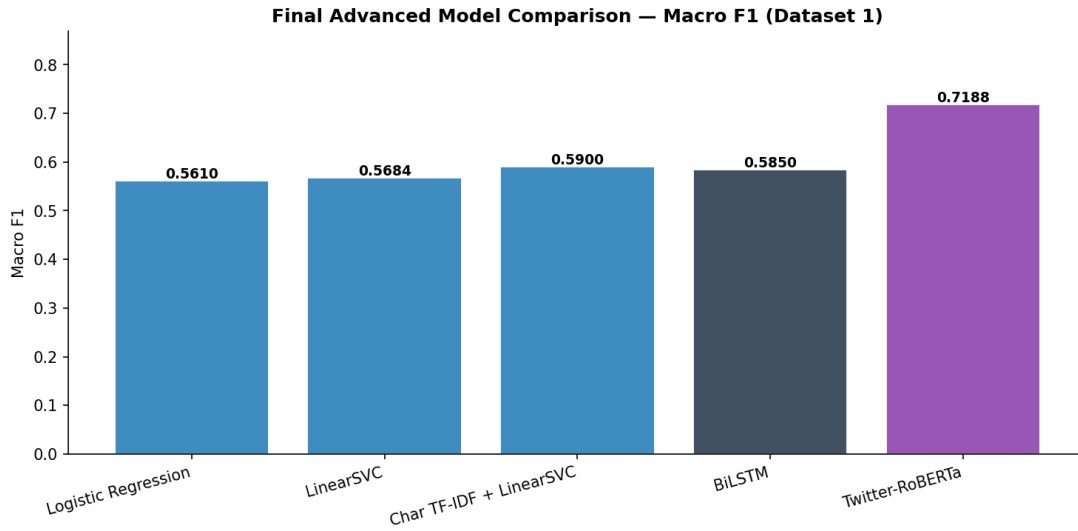


Figure 7.1 — Dataset 1 final advanced-model Macro F1 comparison.

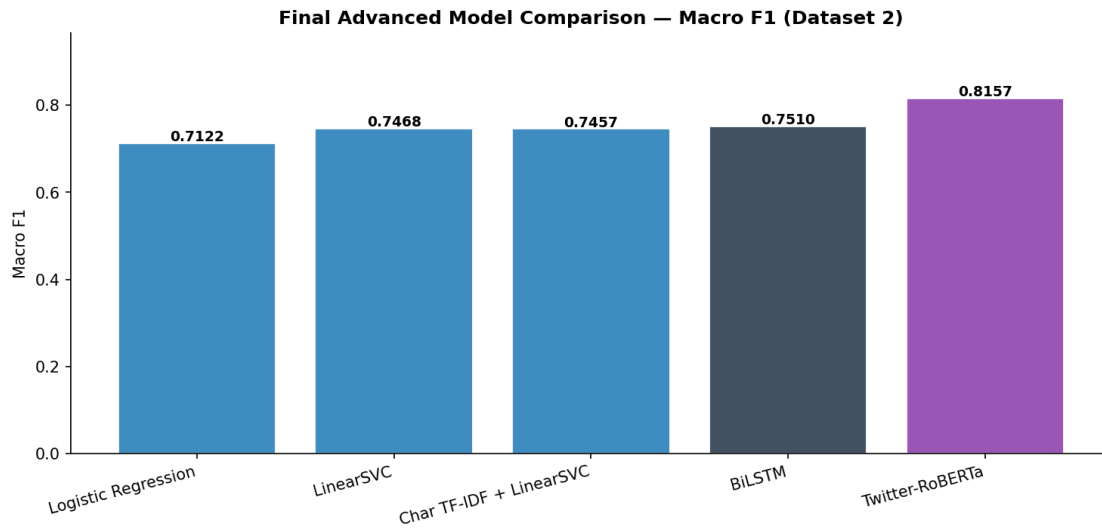


Figure 7.2 — Dataset 2 final advanced-model Macro F1 comparison.

7.2 Dataset 1 class-wise evaluation

Table 7.3 — Dataset 1 LinearSVC class-wise test metrics

Class / aggregate	Precision	Recall	F1	Support
negative	0.6414	0.4215	0.5087	3972.0
neutral	0.6013	0.6729	0.6351	5937.0
positive	0.5007	0.6387	0.5613	2375.0

For Dataset 1 LinearSVC, neutral is the strongest class by F1 at 0.6351, while negative is the weakest at 0.5087. Negative recall is only 0.4215, meaning many genuinely negative tweets are missed. The confusion matrix shows 1,448 negative tweets predicted as neutral and 1,418 neutral tweets predicted as negative. This reciprocal confusion indicates that sparse lexical evidence often does not cleanly separate neutral factual statements from negative language.

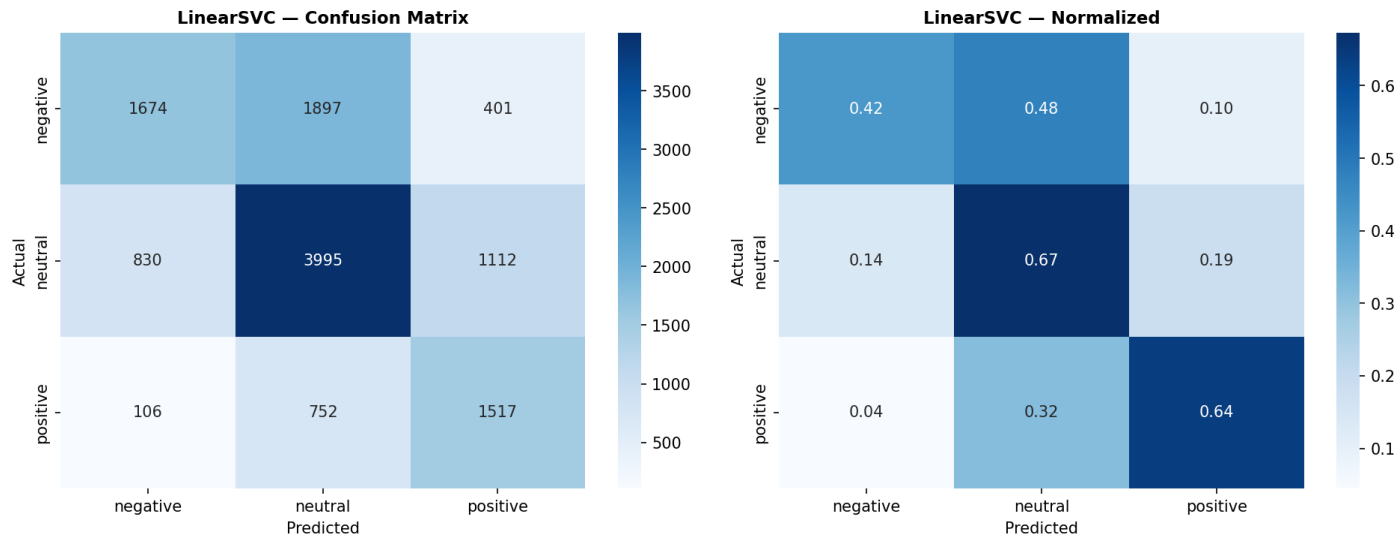


Figure 7.3 — Dataset 1 LinearSVC count confusion matrix.

Figure 7.4 — Dataset 1 LinearSVC row-normalized confusion matrix.

7.3 Dataset 2 class-wise evaluation

Table 7.4 — Dataset 2 LinearSVC class-wise test metrics

Class / aggregate	Precision	Recall	F1	Support
negative	0.8385	0.9354	0.8843	1377.0
neutral	0.6960	0.5957	0.6419	465.0
positive	0.8397	0.6215	0.7143	354.0

On Dataset 2, negative is the strongest LinearSVC class with F1 0.8843 and recall 0.9354. Neutral is the weakest with F1 0.6419 and recall 0.5957. The classifier therefore benefits from the strong lexical signal associated with negative service complaints, while neutral tweets remain harder to distinguish from adjacent sentiment categories.

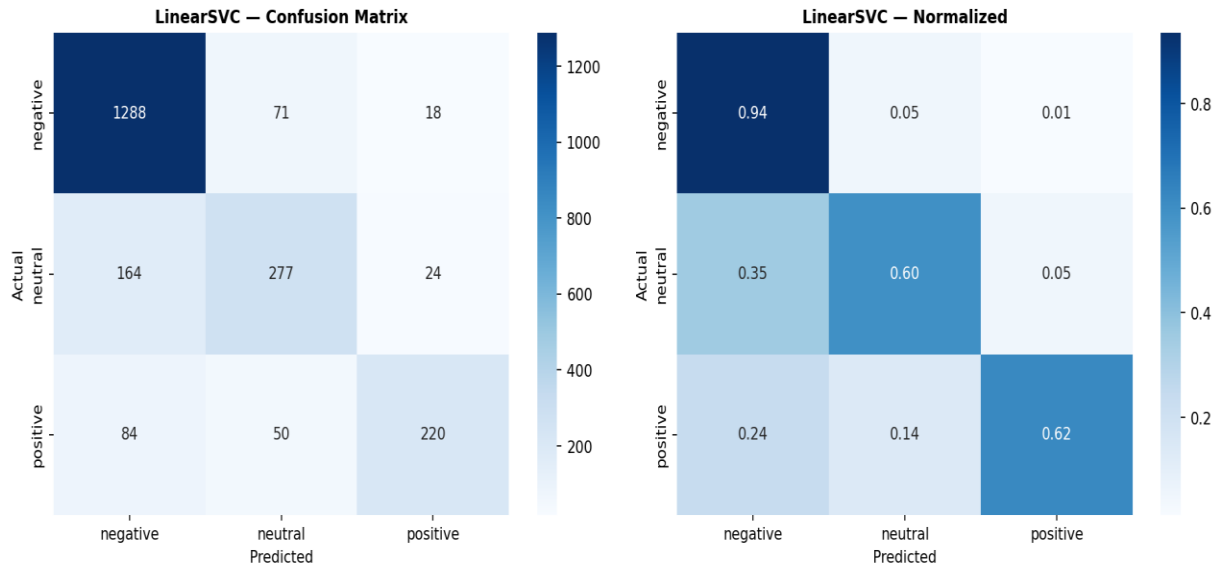


Figure 7.5 — Dataset 2 LinearSVC count confusion matrix.

Figure 7.6 — Dataset 2 LinearSVC row-normalized confusion matrix.

7.4 Twitter-RoBERTa class-wise evaluation

Table 7.5 — Twitter-RoBERTa class-wise metrics derived from the stored confusion matrices

Dataset	Class	Precision	Recall	F1	Support
Dataset 1	negative	0.7089	0.7792	0.7424	3972
Dataset 1	neutral	0.7499	0.6574	0.7006	5937
Dataset 1	positive	0.6690	0.7642	0.7134	2375
Dataset 2	negative	0.9277	0.9136	0.9206	1377
Dataset 2	neutral	0.7149	0.7441	0.7292	465
Dataset 2	positive	0.7949	0.7994	0.7972	354

The stored Twitter-RoBERTa confusion matrices show materially better class separation than the sparse baseline. On Dataset 1, negative recall rises to about 0.779 and positive recall to about 0.764, while neutral remains the hardest class at about 0.657 recall. On Dataset 2, all three recalls are above 0.74, with negative recall near 0.914 and positive recall near 0.799.

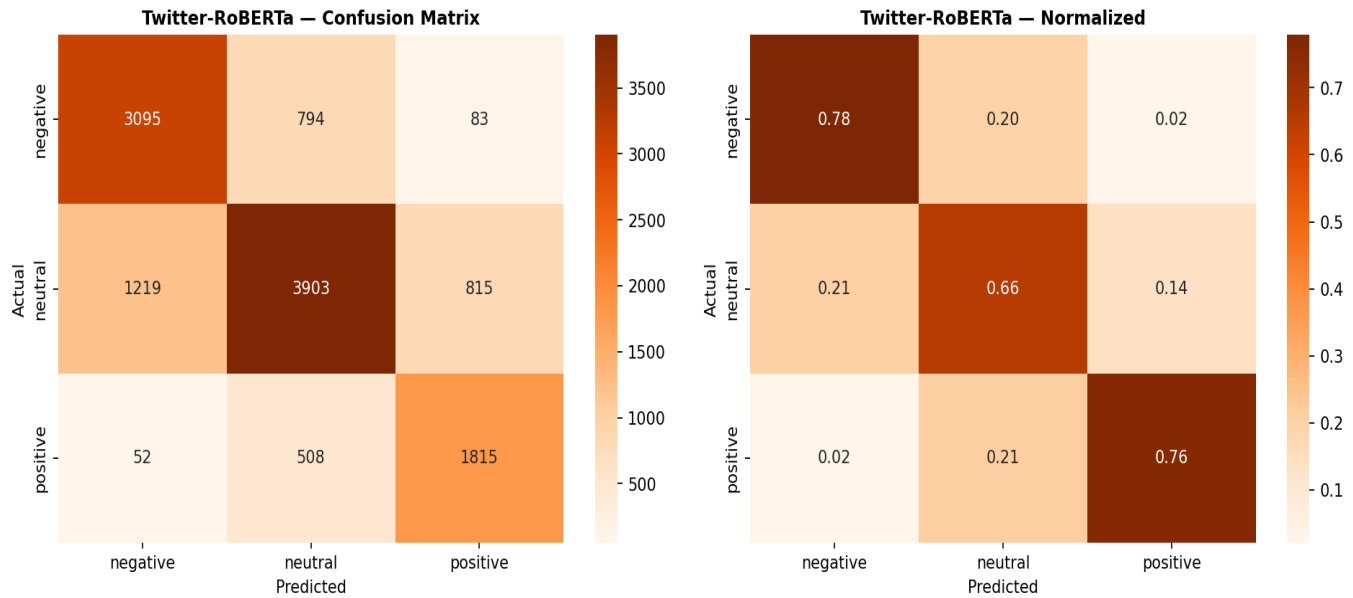


Figure 7.7 — Dataset 1 Twitter-RoBERTa count and normalized confusion matrices.

Figure 7.8 — Dataset 2 Twitter-RoBERTa count and normalized confusion matrices.

8. Interpretation

8.1 Why the classical models behave differently

The classical results are consistent with the geometry of sparse tweet text. TF-IDF creates a high-dimensional representation in which informative words and short phrases can separate sentiment efficiently. LinearSVC is particularly effective because the large-margin objective can exploit sparse discriminative features without requiring a dense representation. Logistic Regression is competitive but slightly weaker in Macro F1 on both datasets. MultinomialNB is fast and useful as a probabilistic reference, but its conditional-independence assumption is restrictive for phrases whose meaning depends on context or word order.

VADER is substantially better than the trivial Dummy baseline on both datasets, confirming that social-media lexical heuristics capture real signal. Nevertheless, its Macro F1 remains below learned TF-IDF models. The gap is expected because VADER does not learn the domain-specific associations present in the training corpus. In the airline data, for example, service terminology and airline-specific complaint patterns can be predictive even when individual words are not globally negative.

8.2 Why the BiLSTM helps only modestly

The BiLSTM improves over the strongest classical word model on Dataset 2, reaching Macro F1 0.7510 versus LinearSVC's 0.7468, but the difference is small. On Dataset 1, the BiLSTM reaches 0.5850 and is only slightly above the word LinearSVC while remaining below the character TF-IDF result. The model is trained from scratch on relatively short sequences, so its representation must learn useful lexical semantics from the available labeled corpus. A pretrained Transformer begins with a much richer contextual representation and therefore has a substantial advantage.

8.3 Error analysis

Table 8.1 — Principal stored error directions

Dataset/model artifact	Actual → predicted	Count	Interpretation
D1 BiLSTM	negative→neutral	1,448	Largest single error direction
D1 BiLSTM	neutral→negative	1,418	Strong reciprocal confusion
D1 BiLSTM	positive→neutral	808	Positive statements diluted into neutral
D2 BiLSTM	neutral→negative	123	Largest single error direction
D2 BiLSTM	negative→neutral	101	Second-largest error direction
D2 BiLSTM	positive→negative	72	Positive service statements misread as complaints

The stored error examples illustrate the linguistic difficulty. In Dataset 2, examples such as “I can legitimately say that I would have rather driven cross country than flown on US Airways” are labeled negative but can be interpreted as neutral if a model underweights the evaluative comparison. Another example, “checking into a flight on your website is about as easy as asking a bear to ... #justsaying,” is neutral in the stored label but strongly negative in lexical form, making it a difficult contextual case. A positive example thanking an airline for service was also misclassified as negative by the BiLSTM, showing that isolated positive words do not guarantee correct polarity when the surrounding service context is complex.

Dataset 1 errors show a different challenge: the corpus is broader and contains political, cultural and topical language. Examples involving hashtags, public figures, quotations and factual statements can be difficult to classify because the sentiment may depend on discourse context rather than an explicit polarity word.

8.4 Linguistic challenge subsets

Table 8.2 — Dataset 1 heuristic challenge-set results

Challenge Type	Count	LinearSVC Accuracy	LinearSVC Macro F1	BiLSTM Accuracy	BiLSTM Macro F1
negation	1296	0.5586	0.5344	0.5602	0.523
emoji_heavy	6025	0.5892	0.5809	0.6073	0.6036
hashtag_heavy	2632	0.6003	0.5546	0.5889	0.5594
short_ambiguous	286	0.6538	0.5802	0.6434	0.5812
slang	164	0.5183	0.5117	0.6159	0.609

Table 8.3 — Dataset 2 heuristic challenge-set results

Challenge Type	Count	LinearSVC Accuracy	LinearSVC Macro F1	BiLSTM Accuracy	BiLSTM Macro F1
emoji_heavy	2196	0.8128	0.7468	0.8097	0.751
negation	444	0.9009	0.6358	0.8874	0.6289
hashtag_heavy	98	0.7347	0.6912	0.7857	0.7456
short_ambiguous	103	0.7573	0.7347	0.767	0.7235
slang	16	0.4375	0.4158	0.5	0.4966

The challenge-set analysis is useful but should be interpreted as a heuristic stress test rather than a clean benchmark. Challenge types can overlap, and the notebook's emoji heuristic is broad. In Dataset 2, the generated challenge table labels all 2,196 test examples as emoji-heavy, so this category is not equivalent to a manually verified emoji-only subset. The challenge results nevertheless show that slang is difficult: Dataset 2 LinearSVC Macro F1 is only 0.4158 and BiLSTM is 0.4966 on the small 16-example slang subset.

8.5 Confidence and abstention

Table 8.4 — Validation-selected confidence abstention

Dataset	Threshold	Total Test Samples	Auto-Accepted	Human-Review	Coverage (%)	Auto-Accepted Accuracy
D1	0.95	12284	178	12106	1.4	0.9438
D2	0.95	2196	237	1959	10.8	0.9873

The abstention rule was selected from validation data and then frozen for the test set. At threshold 0.95, Dataset 1 accepts only 178 of 12,284 test examples automatically, giving 1.4% coverage and 94.38% accuracy among accepted cases. Dataset 2 accepts 237 of 2,196 examples, giving 10.8% coverage and 98.73% accuracy among accepted cases. The high review rates show that a strict confidence policy can improve the reliability of automatic predictions, but only by routing most cases to human review.

9. Subgroup and Robustness

9.1 Airline subgroup analysis

Table 9.1 — Dataset 2 airline sentiment distribution with support

airline	N	Negative %	Neutral %	Positive %
American	2759	71.0	16.8	12.2
Delta	2222	43.0	32.5	24.5
Southwest	2420	49.0	27.4	23.6
US Airways	2913	77.7	13.1	9.2
United	3822	68.9	18.2	12.9
Virgin America	504	35.9	33.9	30.2

The airline-level distribution is highly uneven. United has the largest support at 3,822 tweets, while Virgin America has 504. Negative sentiment ranges from 35.9% for Virgin America to 77.7% for US Airways. American is also strongly negative-heavy at 71.0%. These differences are descriptive properties of the sampled corpus and should not be interpreted as causal measures of service quality or as market-wide customer satisfaction.

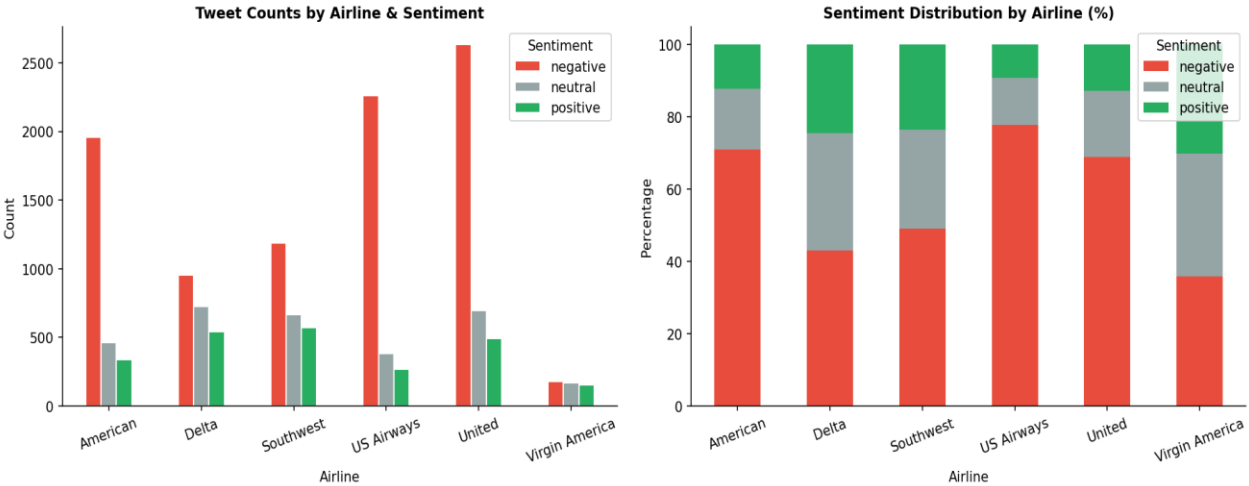


Figure 9.1 — Dataset 2 sentiment distribution by airline.

Figure 9.2 — Dataset 2 negative-tweet rate by airline.

9.2 Cross-dataset transfer

Table 9.2 — Cross-dataset LinearSVC transfer

Transfer	Macro F1	Weighted F1	Accuracy	Interpretation
D1 → D1	0.5684	0.5799	0.5850	Reference within-domain
D1 → D2	0.5338	0.5401	0.5287	Drop of 0.0346 Macro F1
D2 → D2	0.7468	0.8056	0.8128	Reference within-domain
D2 → D1	0.4246	0.4243	0.4553	Drop of 0.3222 Macro F1

The transfer experiment demonstrates asymmetric domain shift. Training on the airline corpus and testing on TweetEval is particularly difficult: Macro F1 drops from 0.7468 within the airline domain to 0.4246 on TweetEval. The reverse transfer is less damaging in absolute terms, but still declines from 0.5684 to 0.5338. The difference is consistent with the airline model learning service-specific lexical patterns that do not generalize to heterogeneous Twitter sentiment.

9.3 Temporal robustness

Table 9.3 — Dataset 2 random versus temporal LinearSVC evaluation

Split	Model	Macro F1	Weighted F1	Accuracy	Macro F1 Change
Random (70/15/15 stratified)	LinearSVC	0.7468	0.8056	0.8128	0.0
Temporal (80% early / 20% late)	LinearSVC	0.7138	0.7632	0.7763	-0.033

The chronological experiment creates an 80% earlier / 20% later split. LinearSVC Macro F1 falls from 0.7468 on the random stratified test to 0.7138 on the later temporal test, with weighted F1 decreasing from 0.8056 to 0.7632 and accuracy from 0.8128 to 0.7763. The Macro F1 reduction of 0.0330 is evidence of temporal drift in this historical corpus. It does not prove a particular causal source of the drift, but it does show that random splitting can be more optimistic than chronological evaluation.

9.4 Multi-seed uncertainty

Table 9.4 — Advanced uncertainty evidence

Model	Mean Macro F1	SD	95% CI	Evidence
Dataset 1 BiLSTM	0.5799	0.0106	0.5477–0.6121	3 completed seeds
Dataset 2 BiLSTM	0.7434	0.0085	0.7176–0.7692	3 completed seeds

The BiLSTM results provide three-seed stability evidence.

10. Limitations and Risk

10.1 Dataset representativeness

Neither dataset represents the full customer population. TweetEval is a benchmark assembled from Twitter tasks rather than a probability sample of customers. The airline dataset is historical, concentrated on six airlines and collected in February 2015. Consequently, the learned vocabulary and sentiment patterns may not reflect current social-media language, current airline operations, or users who do not post publicly.

10.2 Annotation and semantic limitations

Sentiment labels are annotations rather than objective ground truth. Neutral sentiment can be especially ambiguous because a tweet may be factual on the surface while implying dissatisfaction or praise. Sarcasm, irony, mixed sentiment, quotations and domain-specific language can reverse the apparent polarity of individual words. The stored error cases demonstrate this problem.

10.3 Privacy and responsible-use risks

Tweets can contain usernames, locations, URLs and other identifying information. The modeling pipeline replaces usernames and URLs in text, and the report avoids reproducing raw handles. Sentiment predictions should not be used to infer sensitive traits or to make punitive decisions about individuals. Aggregate reporting should disclose support sizes, sampling limitations and uncertainty.

10.4 Leakage risk

The experimental design explicitly separates target labels from annotation metadata and fits TF-IDF within model pipelines. The locked-test principle is followed for the main model comparison. Cross-dataset experiments are also separated from the within-dataset test results. These controls reduce, but do not eliminate, risks associated with duplicate or near-duplicate tweets and domain-specific artifacts.

11. Deployment Considerations

11.1 Appropriate deployment scenario

A responsible deployment would use the classifier as an aggregate monitoring component. New tweets could be classified, grouped by product/service entity and time window, and summarized as sentiment trends. Individual tweets requiring action should be reviewed by a human rather than treated as automatically verified complaints.

11.2 Performance versus efficiency

Table 11.1 — Predictive performance versus computational cost

Model / dataset	Macro F1	Training time	Inference time	Size / scale	Decision value
Dataset 1 LinearSVC	0.5684	7.3 s	0.617 s	Small sparse pipeline	Fast, interpretable baseline
Dataset 1 BiLSTM	0.5850	104.0 s	1.133 s	30.57 MB	Small quality gain over word SVC
Dataset 1 Twitter- RoBERTa	0.7188	1079.5 s	22.84 s	2860.23 MB	Largest quality gain, highest cost
Dataset 2 LinearSVC	0.7468	0.7 s	0.059 s	Small sparse pipeline	Very strong efficiency
Dataset 2 BiLSTM	0.7510	20.4 s	0.592 s	23.25 MB	Marginal quality gain
Dataset 2 Twitter- RoBERTa	0.8157	355.2 s	4.61 s	2860.19 MB	Strongest quality, higher cost

The efficiency comparison changes the deployment recommendation. If the application requires very low latency, transparent behavior and inexpensive retraining, LinearSVC is an attractive baseline, particularly on the airline domain where its Macro F1 is already 0.7468. If the application places a high value on reducing contextual sentiment errors and can support GPU-based training and larger model storage, Twitter-RoBERTa provides a substantial predictive gain.

11.3 Monitoring and retraining

A deployed system should monitor class proportions, confidence distributions, vocabulary drift, entity coverage and periodic human-reviewed accuracy. Temporal evaluation in Dataset 2 shows why this matters: the same model lost 0.0330 Macro F1 under a chronological holdout. Retraining should therefore be triggered by meaningful drift rather than a fixed assumption that the historical model remains valid indefinitely.

11.4 Human review and abstention

The validation-selected 0.95 confidence threshold demonstrates a practical review mechanism. It produces high accuracy among automatically accepted examples, but at the cost of sending most tweets to review. A real system would need a business-defined coverage target and a validation-based threshold chosen jointly with human-review capacity.

11.5 Reproducibility

The notebooks record random seed 42, Python 3.13.15, TensorFlow 2.20.0 and GPU execution. Saved artifacts include classical pipelines, BiLSTM models and tokenizer files, Transformer checkpoints and CSV/figure outputs. Classical and BiLSTM reload checks passed on fixed samples.

12. Conclusion

The completed laboratory establishes a reproducible sentiment-analysis pipeline spanning transparent lexical baselines, sparse classical machine learning, sequence modeling and a Twitter-domain Transformer. The evidence confirms that classical TF-IDF models are strong, fast and interpretable baselines, with LinearSVC consistently leading the learned classical models under cross-validation. Character TF-IDF provides additional value on TweetEval but not a material test-set gain on the airline corpus.

The BiLSTM provides only a modest improvement over LinearSVC on Dataset 2 and does not outperform the character TF-IDF model on Dataset 1. Its three-seed results demonstrate measurable variance, reinforcing the need for uncertainty analysis when drawing comparative conclusions from neural models.

Twitter-RoBERTa is the strongest completed model in point-estimate terms. It reaches Macro F1 0.7188 on TweetEval and 0.8157 on the airline dataset. The improvement over LinearSVC is substantial, particularly on the heterogeneous TweetEval benchmark. However, the gain comes with a major computational cost: roughly 124.6 million parameters, a checkpoint around 2.86 GB and considerably longer training and inference times.

Robustness experiments reinforce the importance of context. Cross-dataset transfer produces large performance degradation, especially from the airline domain to general TweetEval sentiment. Temporal evaluation also reduces performance. Airline-level sentiment distributions vary considerably and must be reported with support rather than interpreted as causal evidence about service quality.

The most defensible recommendation is therefore a two-tier analytical strategy. LinearSVC should remain the default reference model when efficiency, reproducibility and interpretability dominate. Twitter-RoBERTa is justified when the additional contextual accuracy is operationally valuable and the organization can support its computational and reproducibility requirements. In both cases, deployment should remain aggregate and human-supervised, with drift monitoring, confidence-aware review, privacy controls and periodic revalidation.

References

1. Barbieri, F., Camacho-Collados, J., Espinosa Anke, L., & Neves, L. (2020). TweetEval: Unified Benchmark and Comparative Evaluation for Tweet Classification. Findings of the Association for Computational Linguistics: EMNLP 2020, 1644–1650. <https://aclanthology.org/2020.findings-emnlp.148/>
2. Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
3. Hutto, C. J., & Gilbert, E. (2014). VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1), 216–225. <https://doi.org/10.1609/icwsm.v8i1.14550>
4. Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., & Stoyanov, V. (2019). RoBERTa: A Robustly Optimized BERT Pretraining Approach. *arXiv:1907.11692*.
5. Faculty laboratory manual: MDI3003 – Advanced Predictive Analytics, Laboratory Experiment 05, Product and Brand Sentiment Prediction from Tweet Data, Fall Semester 2026–2027.
6. CardiffNLP. TweetEval sentiment benchmark and Twitter-domain sentiment model resources.
7. CrowdFlower / Figure Eight. Twitter US Airline Sentiment dataset, distributed through Kaggle; public dataset description states February 2015 collection and positive/negative/neutral annotation.
8. scikit-learn documentation. Text feature extraction, pipelines, model selection, LinearSVC, Logistic Regression, MultinomialNB and classification metrics.
9. Hugging Face Transformers documentation and CardiffNLP model card for cardiffnlp/twitter-roberta-base-sentiment-latest.

LinearSVC is the strongest classical baseline; Twitter-RoBERTa is the strongest completed model; temporal and cross-dataset tests demonstrate measurable domain sensitivity.